

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B.Tech Branch :Electronics and Telecommunication Engineering/ Electronics and Communication Engineering

Subject Code & Name: BTETOE505A Control System Engineering Semester: V

Max Marks: 60

Date: 22/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	In force-voltage analogy, Mass element is equal to _____ a. Resistance b. Inductance c. Capacitance d. Conductance	L1/CO1	1
2	For Impulse input, the output response C(s) is equal to. a. R(s) b. E(s) c. G(s) d. B(s)	L1/CO1	1
3	The position and velocity error constants of a type-2 system are a. Constant, constant b. constant, infinity c. zero, constant d. infinity, infinity	L1/CO2	1
4	Velocity error constant of a system is measured when the input to the system is unit a. Parabolic b. ramp c. impulse d. step	L1/CO2	1
5	In case of type-1 system steady state error for parabolic input is a. Unity b. infinity c. zero d. 10	L1/CO2	1
6	The stability of the system can be increased by adding____ a. Pole b. zero c. both d. none	L1/CO2	1
7	From the bode plots it is observed that the gain cross over frequency is greater than phase cross over frequency. The system is called____ a. Stable b. Marginally stable c. Conditionally stable d. Unstable	L2/CO3	1
8	The slope of $(1+j\omega)$ is a. +20db b. +40db c. -40db d. -20db	L3/CO3	1

9	The function of the proportional plus integral mode is to _____?				L2/CO4	1
	a. Provide gain	b. Eliminate offsets	c. To speed up the response and minimize the overshoot	d. All of the above		
10	The PID controller is very effective for _____ processes?				L1/CO4	1
	a. Slower	b. Faster	c. poor	d. none of these		
11	The smallest set of variable of a state is called				L1/CO5	1
	a. State	b. condition of state	c. Eigen values	d. state variables		
12	State space analysis is applicable even if the initial conditions are				L1/CO5	1
	a. Zero	b. Non-zero	c. Equal	d. Not equal		
Q. 2 Solve the following.						12
A)	For the system represented in the given figure, determine transfer function $C(S)/R(S)$				L3/CO1	6
B)	Find the overall transfer function of the system whose signal flow graph is shown below.				L5/CO1	6
Q. 3 Solve the following.						12
A)	Control system is having an unity feedback and $G(s) = \frac{20}{s(1+4s)(1+s)}$ Determine: 1) Type of System 2) Static Error Coefficients 3) Steady state error if input is $r(t) = 2 + 4t + \frac{t^2}{2}$				L3/CO2	6
B)	Derive and expression for rise time and peak time				L5/CO2	6

Q. 4	Solve Any Two of the following.		12
A)	Determine the stability of $s^5 + s^4 + 2s^3 + 2s^2 + 3s + 5 = 0$	L3/CO2	6
B)	Draw the Bode plot and find gain margin and phase margin for the following Transfer Function $G(s)H(s) = \frac{10}{s(s+1)(s+5)}$	L3/CO3	6
C)	Sketch the polar plot for the open loop transfer function of a unity feedback system is given by $G(s) = \frac{1}{s(1+s)(1+2s)}$ also determine Gain Margin & Phase.	L3/CO3	6
Q.5	Solve Any Two of the following.		12
A)	Differentiate between phase lead and lag compensation	L3/CO4	6
B)	Obtain the transfer function of Lead Compensator, draw pole-zero plot and write the procedure for design of Lead Compensator using Bode plot.	L3/CO4	6
C)	The open loop transfer function of a uncompensated system is $G(S) = \frac{5}{S(S+2)}$ Design a suitable lag compensator for the system so that the static velocity error constant $k_v = 1$ 20 sec, phase margin is at least 55° and the gain margin is at least 12 dB.	L3/CO4	6
Q. 6	Solve Any Two of the following.		12
A)	Obtain the state model for the system with transfer function $\frac{Y(s)}{U(s)} = \frac{3s+4}{s^2+5s+6}$	L3/CO5	6
B)	Obtain State Transition matrix of the system $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 0 & -3 \end{bmatrix} x(t)$	L3/CO5	6
C)	Comment on Controllability and Observability of System $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$	L3/CO5	6
*** End ***			